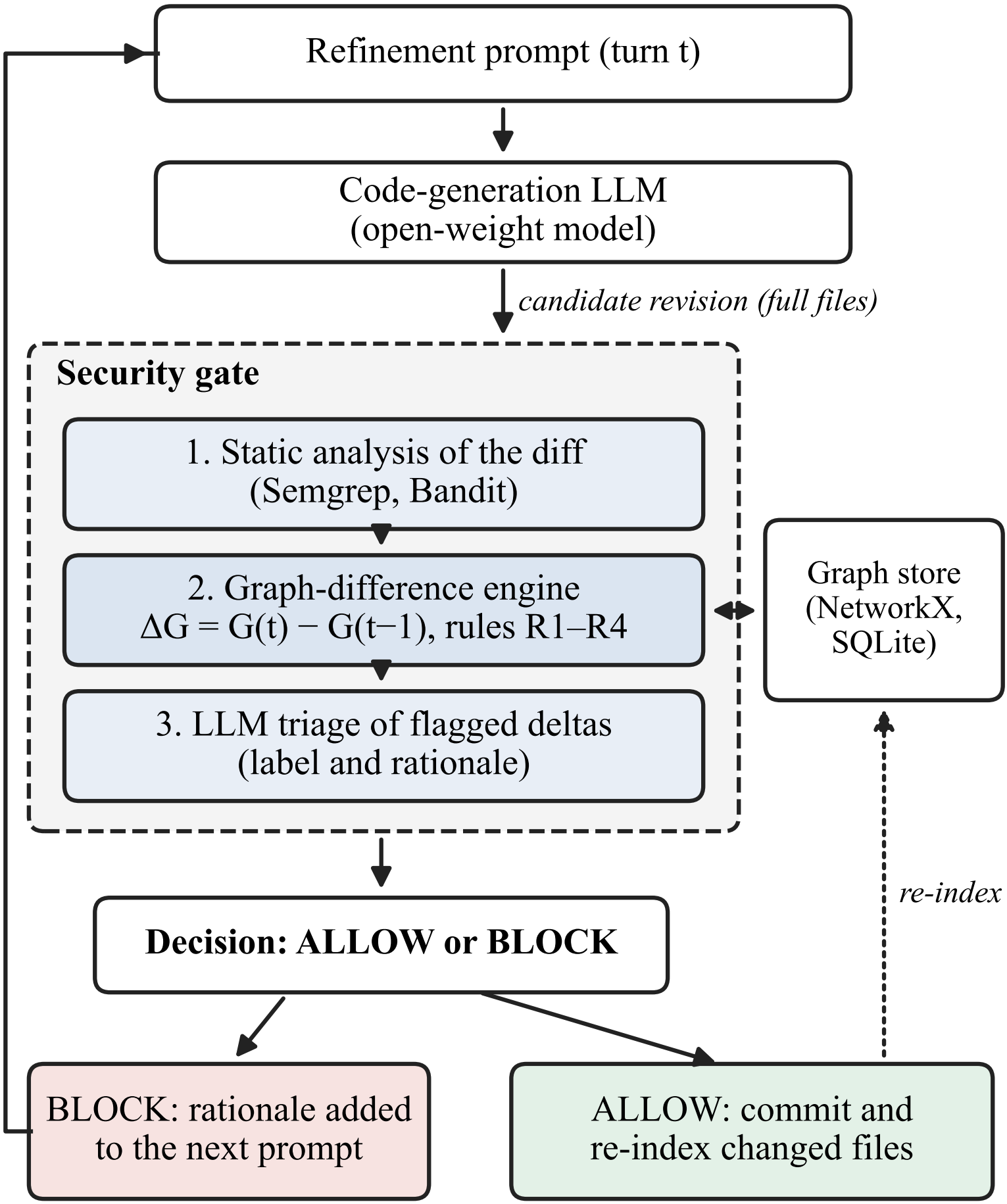
Refinement prompt (turn t)
Code-generation LLM
(open-weight model)
candidate revision (full files)
Security gate
1. Static analysis of the diff
(Semgrep, Bandit)
2. Graph-difference engine
ΔG = G(t) − G(t−1), rules R1–R4
3. LLM triage of flagged deltas
(label and rationale)
Graph store
(NetworkX,
SQLite)
re-index
Decision: ALLOW or BLOCK
BLOCK: rationale added
to the next prompt
ALLOW: commit and
re-index changed files